PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 1 • Chapter OL-T33 Classified

Cross-session • chapter-organised candidate

14 classified items • 41 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Unit 1 • Chapter OL-T33 • 14 items • 41 marks

Chapter	Items	Marks	Starts
01. OL-T33 Graphs of Functions	14	41	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Graphs of Functions

Unit 1 • 41 marks • 14 item(s)

June 2025 | 4MA1-1H Q10 | 3 marks
Plot a cubic from a table

June 2025 | 4MA1-2H Q14 | 2 marks
Recognise polynomial graph shape

June 2025 | 4MA1-2H Q25 | 3 marks
Feature-led polynomial sketch

November 2023 | 4MA1-1H Q12 | 4 marks
Plot a cubic from a table

November 2023 | 4MA1-1H Q12 | 1 marks
Recognise reciprocal or rational graph shape

June 2023 | 4MA1-2H Q4 | 4 marks
Plot a quadratic from a table

June 2023 | 4MA1-2H Q18 | 3 marks
Recognise reciprocal or rational graph shape

November 2024 | 4MA1-1H Q11 | 4 marks
Plot a reciprocal or mixed rational curve

January 2020 | 4MA1-1H Q20 | 4 marks
Feature-led polynomial sketch

January 2020 | 4MA1-2H Q15 | 3 marks
Recognise polynomial graph shape

June 2021 | 4MA1-2H Q15 | 2 marks
Plot a reciprocal or mixed rational curve

June 2021 | 4MA1-2H Q15 | 2 marks
Plot a reciprocal or mixed rational curve

January 2023 | 1H Q14 | 3 marks
Plot a cubic from a table

January 2023 | 2H Q11 | 3 marks
Plot a cubic from a table

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 07

4MA1-1H Q10

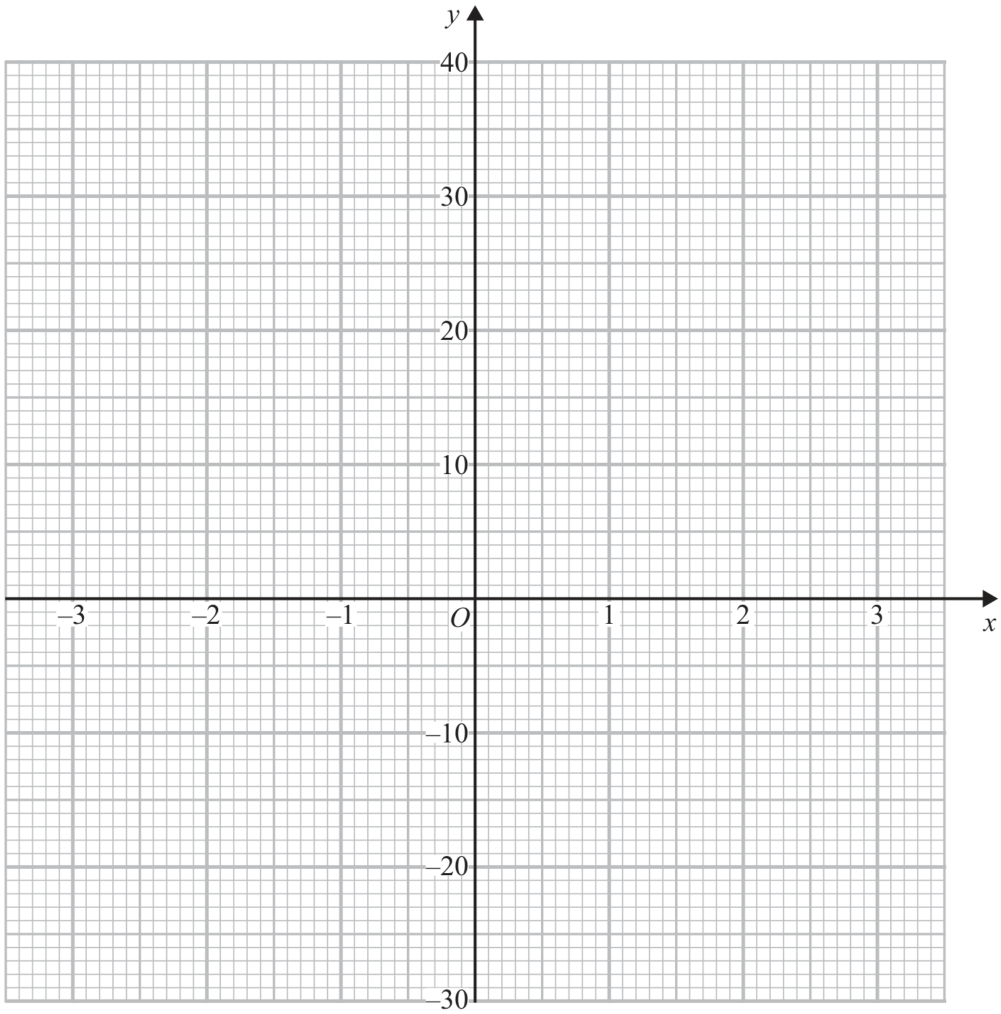
ORIGINAL QUESTION
3 marks

10 (a) Complete the table of values for $y = x^3 + 2x + 3$

x	-3	-2	-1	0	1	2	3
y		-9	0	3	6		36

(1)

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leqslant x \leqslant 3$



(2)

Worked Solution 07

Complete and draw a cubic graph

UNIT 1
3 marks

Complete the table

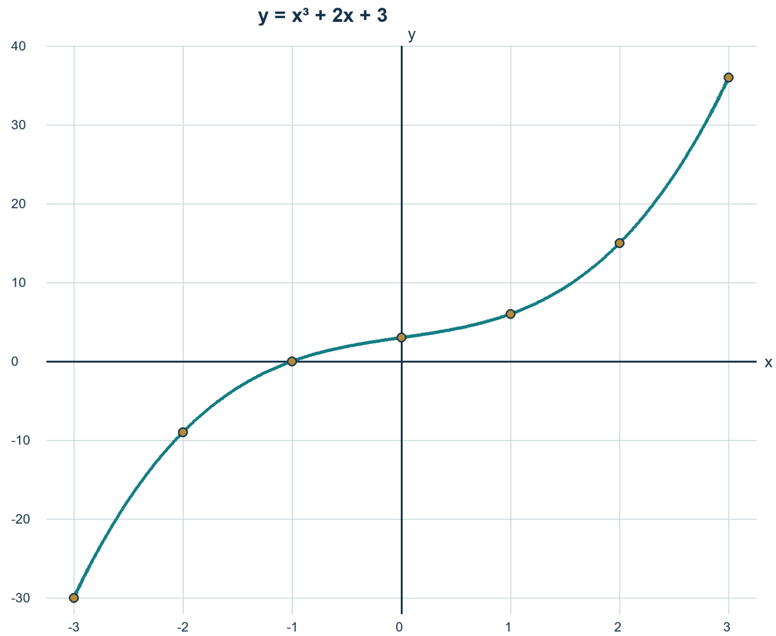
$$y(-3) = -27 - 6 + 3 = -30, \qquad y(2) = 8 + 4 + 3 = 15.$$

Plot

$$(-3, -30), (-2, -9), (-1, 0), (0, 3), (1, 6), (2, 15), (3, 36).$$

Join

$$Draw one smooth increasing cubic curve through all seven points for -3 \le x \le 3.$$



Solved reference diagram

Final answer

$$\text{Missing values } -30 \text{ and } 15; \text{ smooth cubic through all seven points.}$$

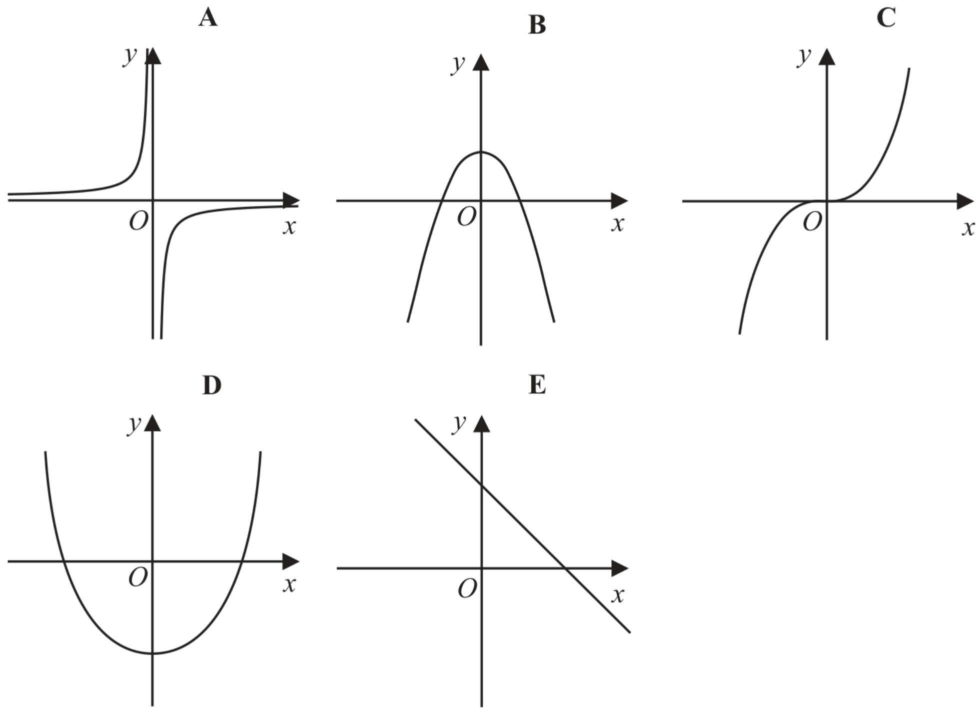
Dependency check The graph uses the completed table; do not use straight-line joins.

Working Unit 20

4MA1-2H Q14

ORIGINAL QUESTION
2 marks

14 Here are five graphs.



(a) Write down the letter of the graph that could have the equation $y = 5 - x^2$

....
(1)

(b) Write down the letter of the graph that could have the equation $y = 2x^3$

....
(1)

Worked Solution 20

Recognise graphs from equations

UNIT 1
2 marks

Part (a)

$y = 5 - x^2$ is a downward parabola with positive y-intercept: graph B.

Part (b)

$y = 2x^3$ is an increasing cubic through the origin: graph C.

Final answer

(a) B (b) C

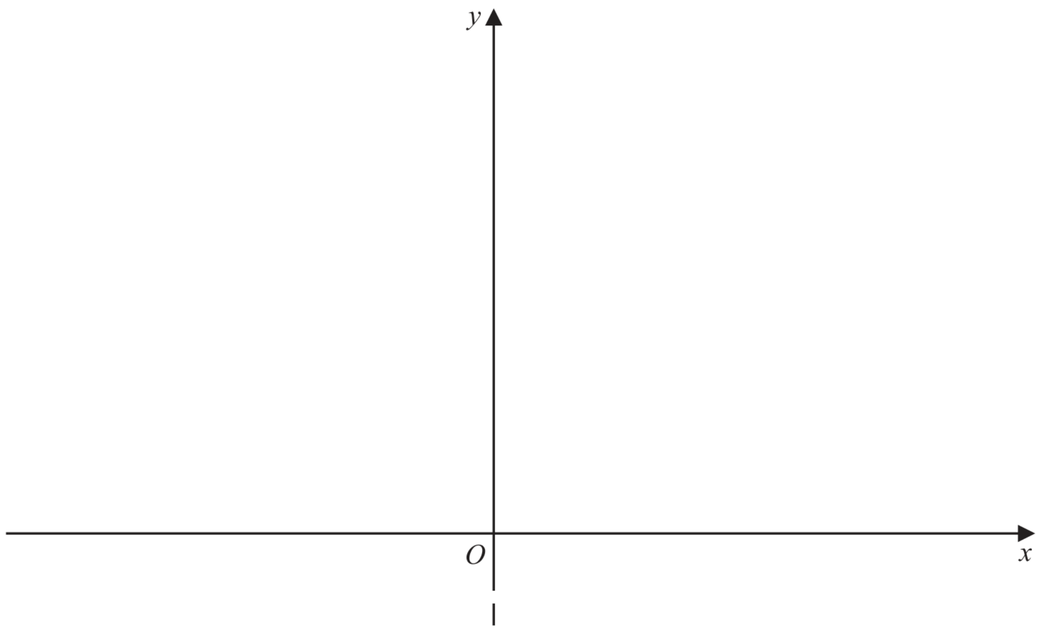
Working Unit 28

4MA1-2H Q25

ORIGINAL QUESTION
3 marks

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y-axis.



(3)

Worked Solution 28

Sketch a quadratic from completed-square form

UNIT 1
3 marks

Turning point

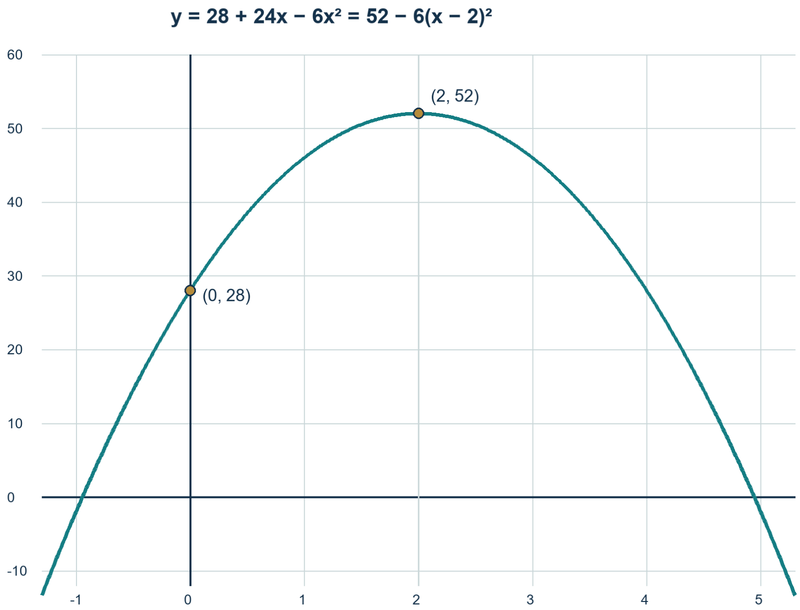
$$y = 52 - 6(x-2)^2 \Rightarrow \text{ maximum turning point } (2, 52).$$

y-intercept

$$x = 0 \Rightarrow y = 28, \text{ so the graph crosses the } y\text{-axis at } (0, 28).$$

Shape

The coefficient of $(x-2)^2$ is negative, so draw a symmetric downward parabola about $x = 2$.



Solved reference diagram

Final answer

Downward parabola with turning point $(2, 52)$ and y-intercept $(0, 28)$.

Dependency check Sequenced immediately after WU027; the printed equation is repeated, so the crop is self-contained.

Question 11

4MA1-1H Q12 (a), (b)

ORIGINAL QUESTION UNIT 1

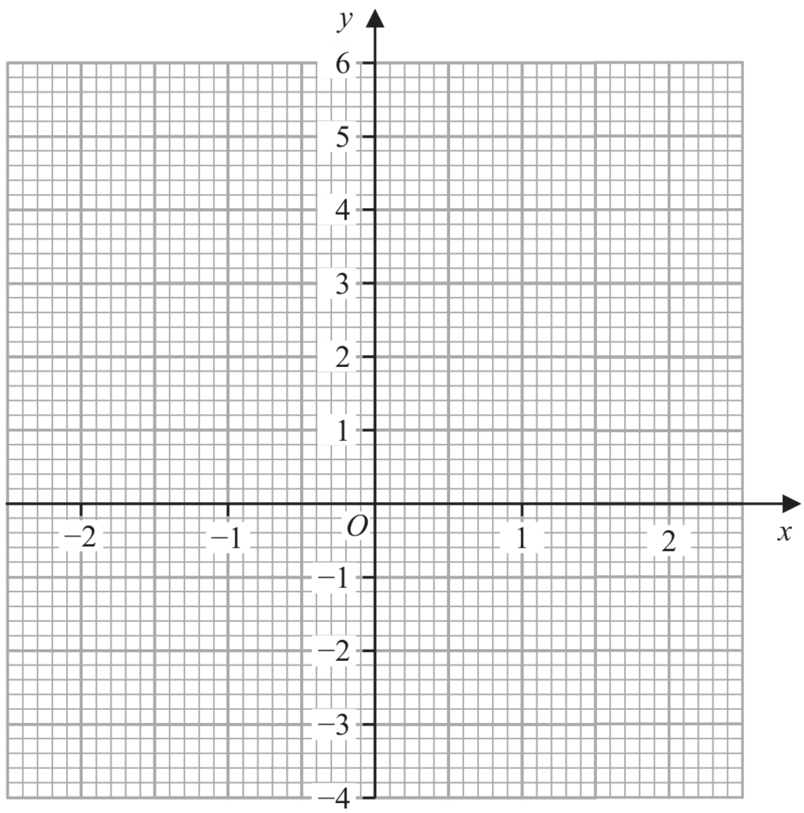
4 marks

12 (a) Complete the table of values for $y = x^3 - 3x + 1$

x	-2	-1	0	1	2
y				-1	

(2)

(b) On the grid, draw the graph of $y = x^3 - 3x + 1$ for values of x from -2 to 2



(2)

Worked solution 11

Complete the cubic table / Draw the cubic

MODULAR UNIT 1
4 marks

(a) – Complete the cubic table

Substitute each x

$$y = x^3 - 3x + 1 \Rightarrow y(-2,-1,0,1,2) = (-1,3,1,-1,3).$$

Final answer

y values: -1, 3, 1, -1, 3

(b) – Draw the cubic

Plot the table points

$$(-2,-1), (-1,3), (0,1), (1,-1), (2,3).$$

Join smoothly

$$Draw one smooth cubic curve through all five points for -2 \le x \le 2.$$

Final answer

Smooth cubic through (-2,-1), (-1,3), (0,1), (1,-1), (2,3)

Completed diagram 11

Complete the cubic table / Draw the cubic

MODULAR UNIT 1

4 marks

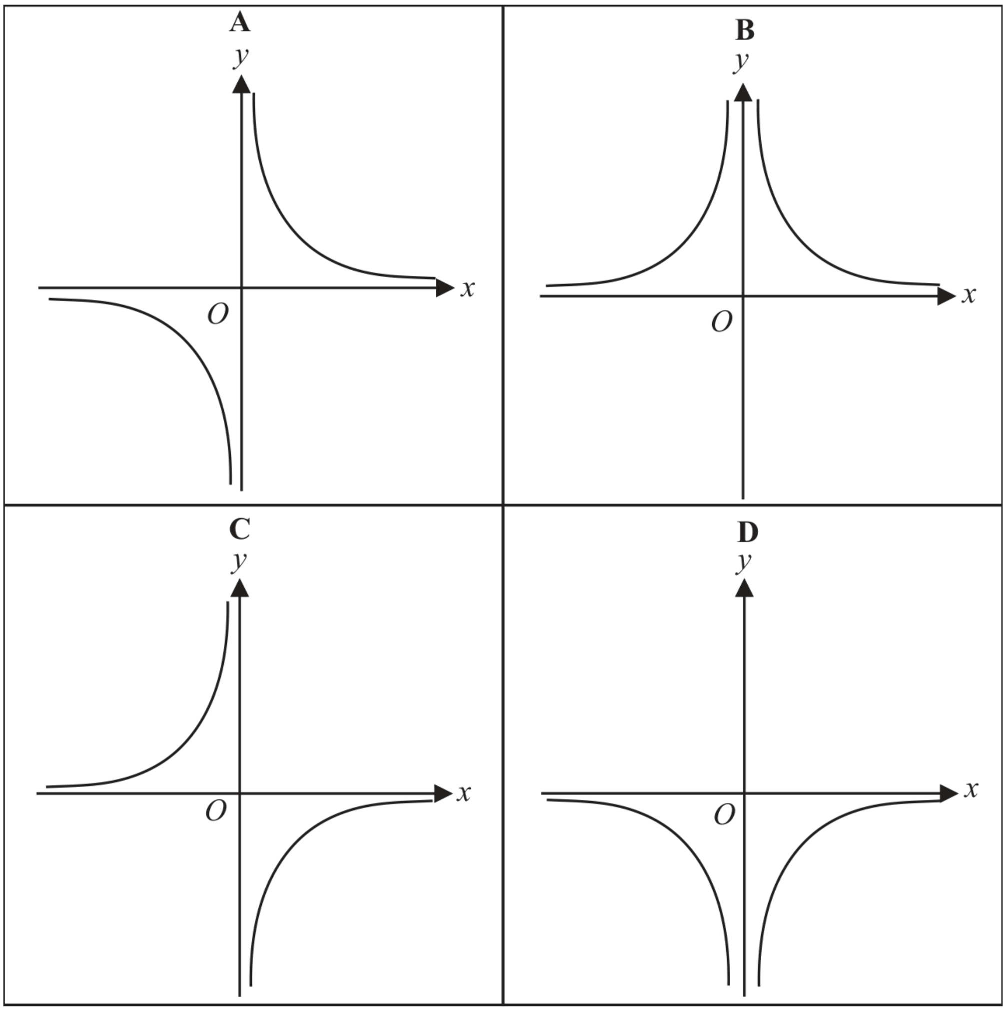
Question 12

4MA1-1H Q12 (c)

ORIGINAL QUESTION UNIT 1

1 marks

Here are four graphs.



(c) Write down the letter of the graph that could have the equation $y = -\frac{1}{x^2}$

....

(1)

Worked solution 12

Recognise y = $-1/x^2$

MODULAR UNIT 1
1 marks

(c) – Recognise y = $-1/x^2$

Use the sign and asymptotes

$$y = -1/x^2 < 0 for x \neq 0, with branches in quadrants III and IV; this is graph D.$$

Final answer

D

Question 15

4MA1-2H Q4

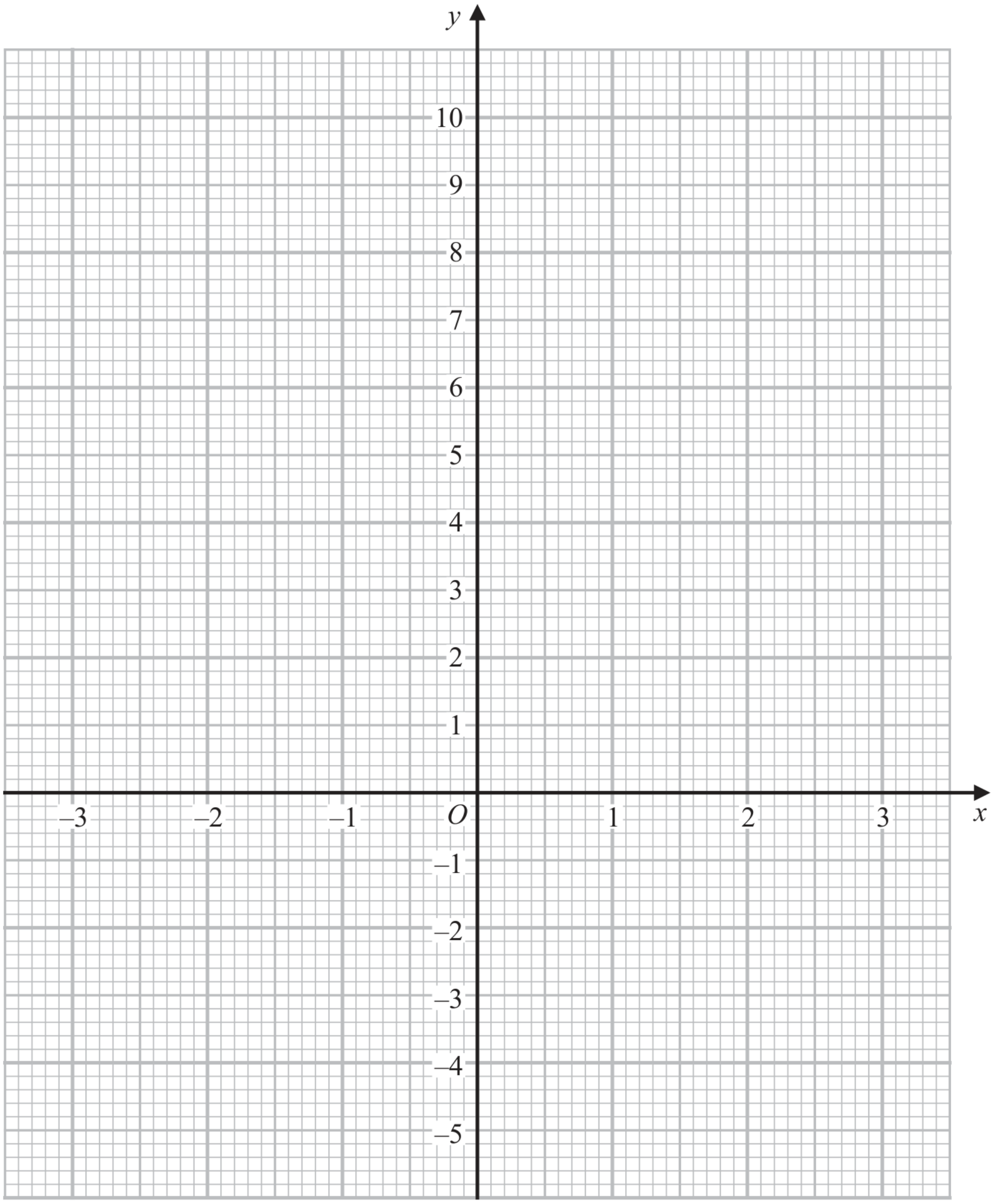
ORIGINAL QUESTION
4 marks

4 (a) Complete the table of values for $y = x^2 - x - 4$

x	-3	-2	-1	0	1	2	3
y		2			-4		

(2)

(b) On the grid below, draw the graph of $y = x^2 - x - 4$ for values of x from -3 to 3



(2)

Worked Solution 15

Complete and draw a quadratic graph

MODULAR UNIT 1

4 marks

Complete the table

$$y = x^2 - x - 4 \Rightarrow (8,\ 2,\ -2,\ -4,\ -4,\ -2,\ 2) \text{ for } x = -3, -2, -1, 0, 1, 2, 3.$$

Plot and join

Plot $(-3, 8), (-2, 2), (-1, -2), (0, -4), (1, -4), (2, -2), (3, 2)$ and join with a smooth quadratic curve.

Final answer

$$y = x^2 - x - 4; \quad y\text{-values } 8, 2, -2, -4, -4, -2, 2$$

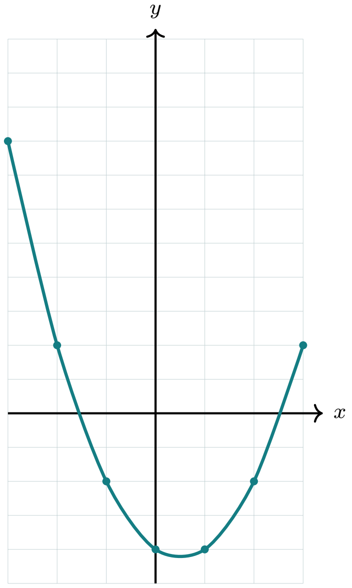
Completed diagram 15

Complete and draw a quadratic graph

MODULAR UNIT 1

4 marks

Completed solution diagram: Completed quadratic graph

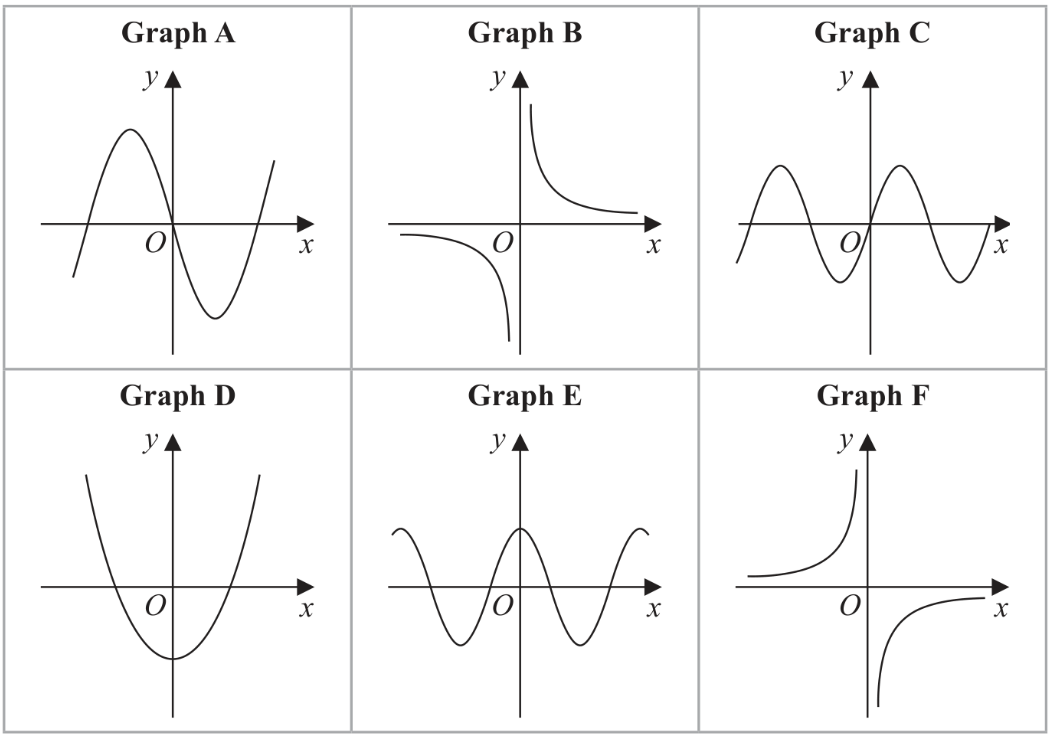


Question 23

4MA1-2H Q18

ORIGINAL QUESTION
3 marks

18 Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \sin x$	....
$y = -\frac{3}{x}$	....
$y = 4x^3 - 5x$	....

Worked Solution 23

Recognise function graph families

MODULAR UNIT 1
3 marks

Match the sine graph

$$y = \sin x \text{ passes through the origin and oscillates : Graph C.}$$

Match the reciprocal graph

$$y = -3/x \text{ has positive branch for } x < 0 \text{ and negative branch for } x > 0 : GraphF.$$

Match the cubic

$$y = 4x^3 - 5x \text{ is an odd cubic with negative central gradient : Graph A.}$$

Final answer

$$\sin x \to C, \quad -3/x \to F, \quad 4x^3 - 5x \to A$$

Original question 08

4MA1-1H Q11 M01, M02

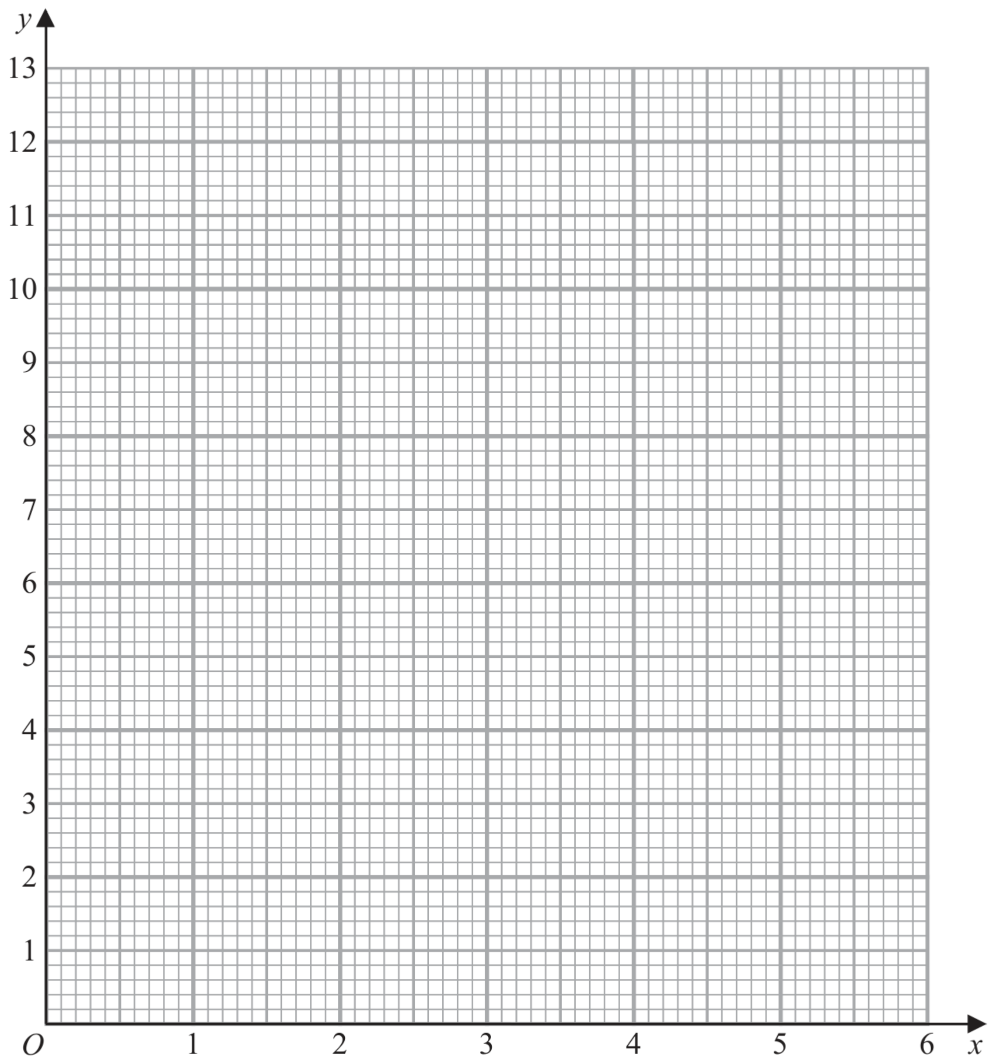
MODULAR UNIT 1
4 marks

11 (a) Complete the table of values for $y = 2\left(x + \frac{1}{x}\right)$

x	0.5	1	2	3	4	5	6
y			5	6.7			12.3

(2)

(b) On the grid, draw the graph of $y = 2\left(x + \frac{1}{x}\right)$ for $0.5 \leqslant x \leqslant 6$



(2)

Worked solution 08

Complete a function table / Draw the function graph

MODULAR UNIT 1
4 marks

M01 – Complete a function table

Use the displayed formula

$$y = 2(x + 1/x).$$

Substitute the four missing x-values

$$y(0.5) = 5,\ y(1) = 4,\ y(4) = 8.5,\ y(5) = 10.4.$$

Final answer

Missing values: 5, 4, 8.5, 10.4

M02 – Draw the function graph

Plot the table

Plot the seven table points for 0.5≤ x≤6.

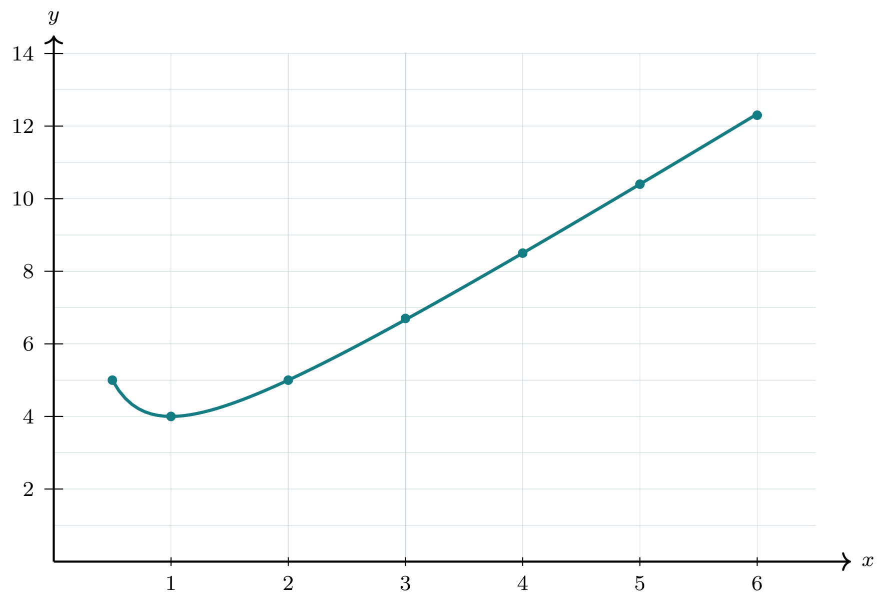
Join smoothly

Join the points with the correct smooth curve on the given grid.

Final answer

The correct smooth curve through the table points

Completed solution diagram: Smooth curve through the table points

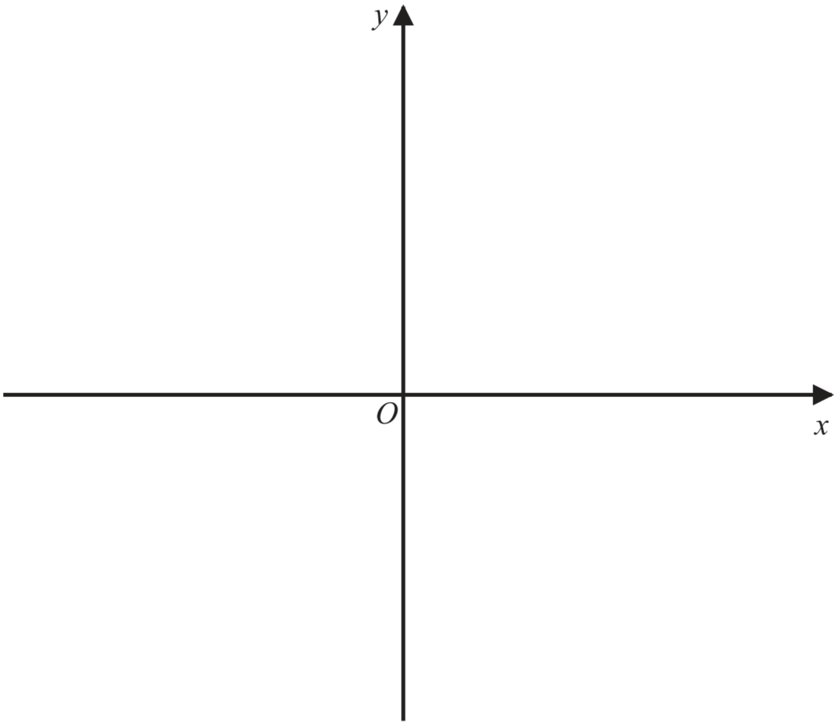


20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of a,

(i) the coordinates of any points of intersection of **C** with the coordinate axes,

(ii) the coordinates of the turning point.



Worked solution

January 2020 | Question 20 | 4 marks

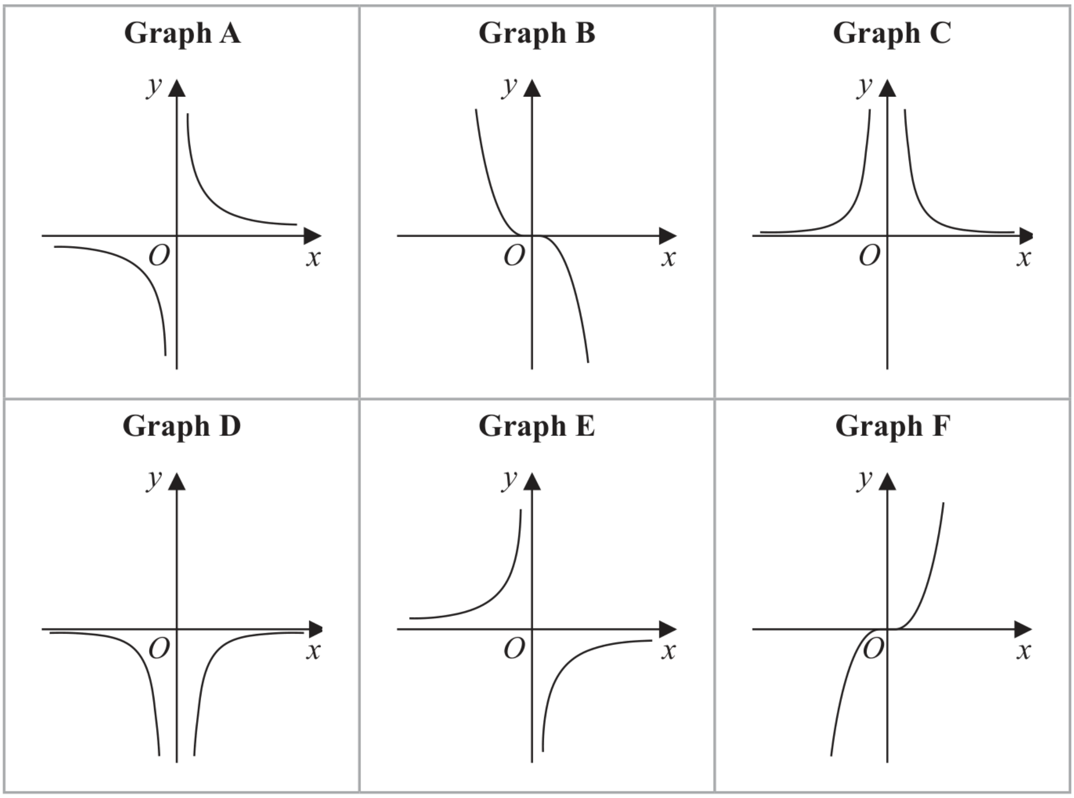
Family: Sketch key features without a full table • Variant: Feature-led polynomial sketch

Method / working

1. Set $y=0$: $4(x-1)^2=a$, so $x=1\pm\sqrt{a}/2$.
2. At $x=0$, $y=4-a$.
3. The vertex form shows turning point $(1,-a)$.
4. Sketch an upward parabola and label these points.

Final answer: Intercepts $(1\pm\sqrt{a}/2,0)$; y-intercept $(0,4-a)$; turning point $(1,-a)$

15 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

Equation	Graph
$y = \frac{2}{x^2}$	....
$y = -\frac{1}{2}x^3$	....
$y = -\frac{5}{x}$	....

Worked solution

January 2020 | Question 15 | 3 marks

Family: Recognise a function graph family • Variant: Recognise polynomial graph shape

Method / working

1. $y=-2/x^2$ is negative and y-axis symmetric: choose graph C.
2. $y=-(1/2)x^3$ is a decreasing cubic through the origin: choose graph B.
3. $y=-5/x$ is a negative reciprocal graph: choose graph E.

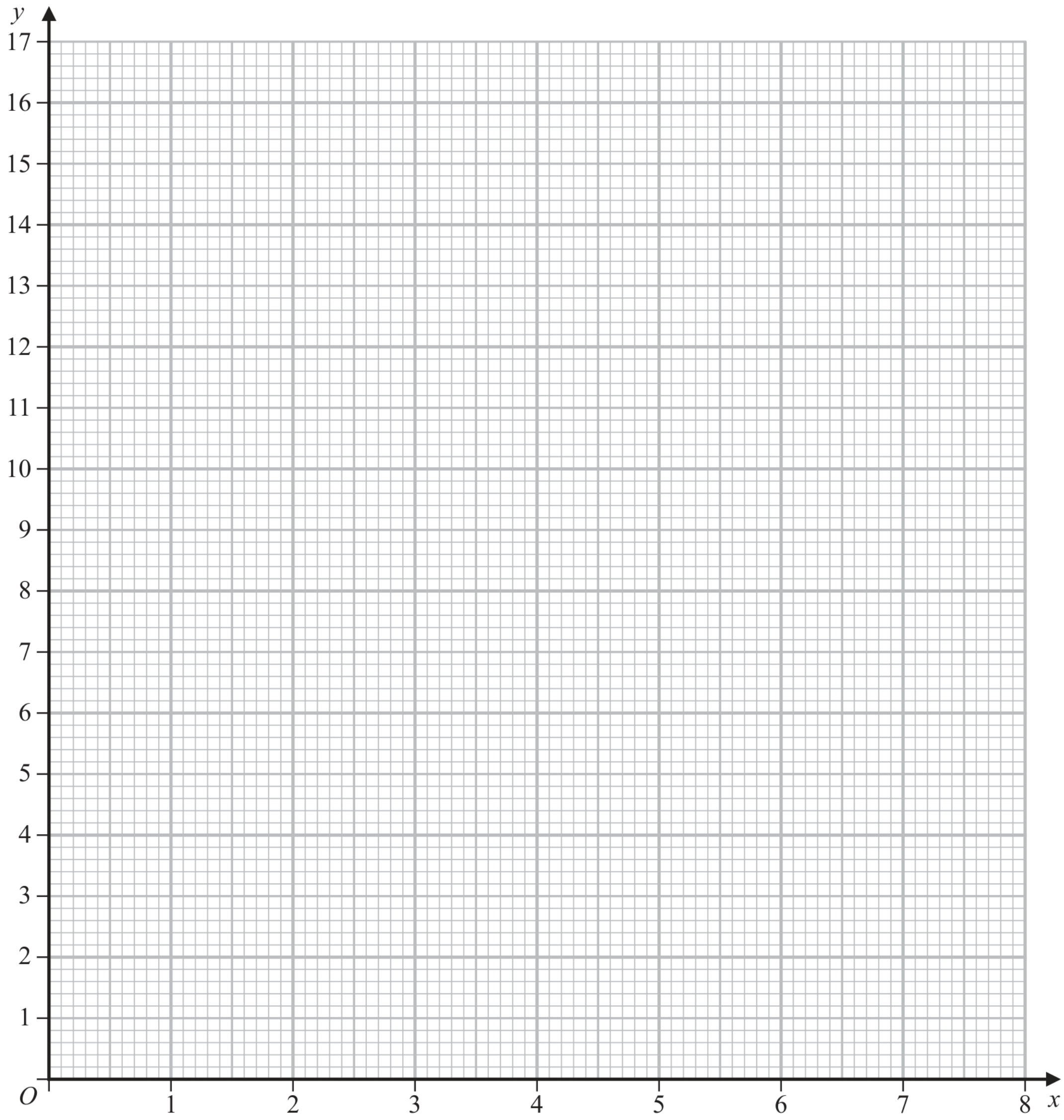
Final answer: C, B, E

15 (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(2)

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leqslant x \leqslant 8$



(2)

Worked solution

June 2021 | Question 15 | 2 marks

Family: Generate a table and draw a nonlinear graph • Variant: Plot a reciprocal or mixed rational curve

Method / working

1. Use y=(x^2+4)/x=x+4/x for x=.5, 1, 2 and 4.
2. The missing values are 8.5, 5, 4 and 5 respectively.

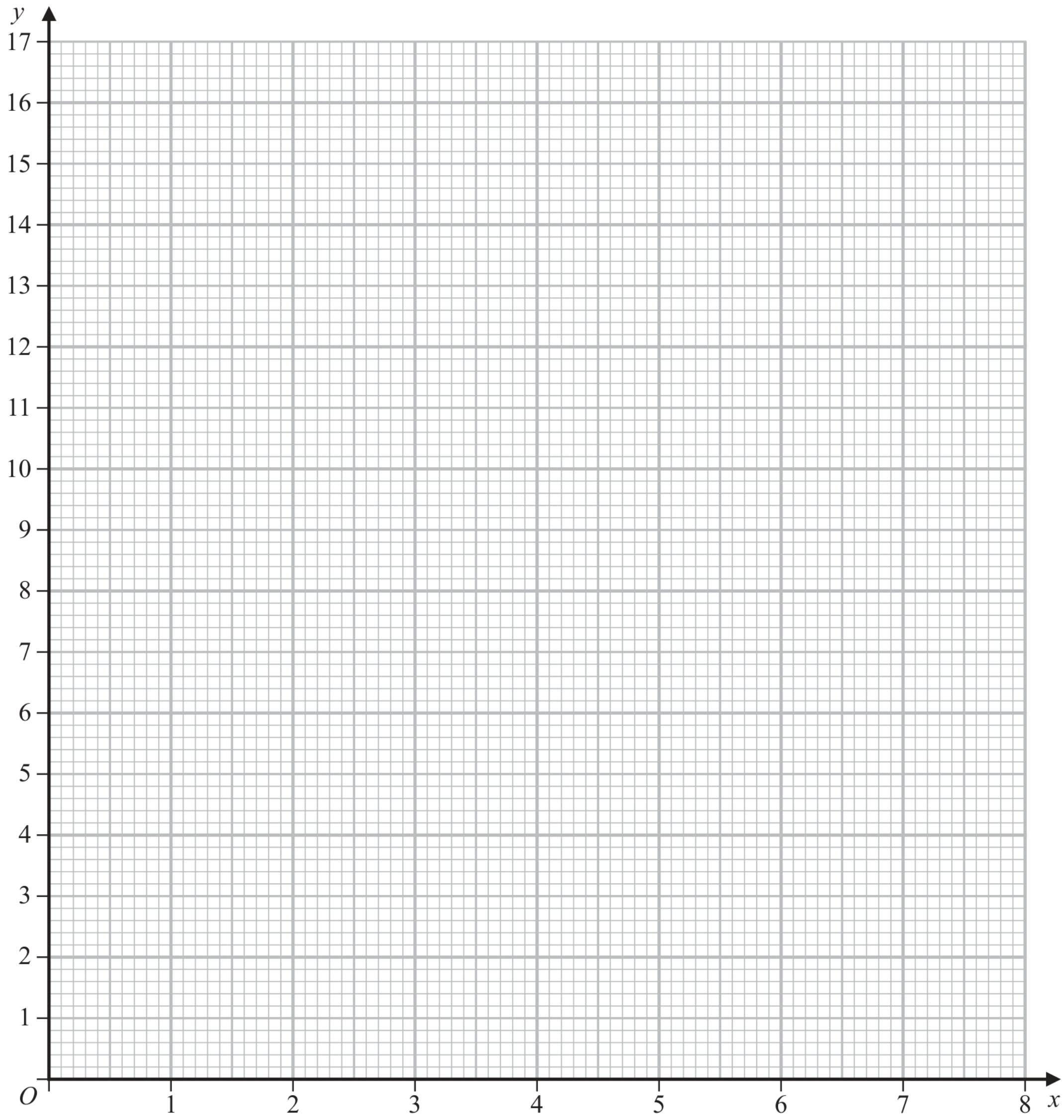
Final answer: 8.5, 5, 4, 5

15 (a) Complete the table of values for $y = \frac{1}{x}(x^2 + 4)$

x	0.25	0.5	1	2	4	8
y	16.25					8.5

(2)

(b) On the grid, draw the graph of $y = \frac{1}{x}(x^2 + 4)$ for $0.25 \leqslant x \leqslant 8$



(2)

Worked solution

June 2021 | Question 15 | 2 marks

Family: Generate a table and draw a nonlinear graph • Variant: Plot a reciprocal or mixed rational curve

Method / working

1. Plot all six points from the completed table accurately on the supplied grid.
2. Join them with a smooth curve for 0.25<=x<=8.

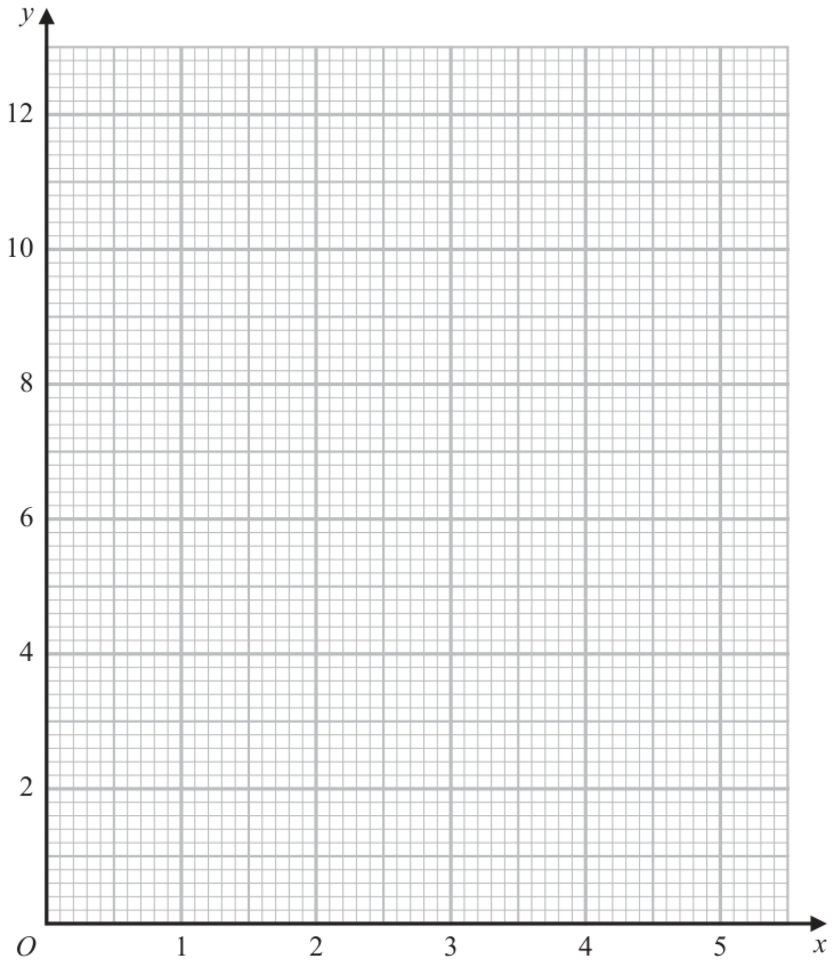
Final answer: Smooth graph through the six table points

14 (a) Complete the table of values for $y = \frac{2}{x}\left(5 - \frac{1}{x}\right)$

x	0.5	1	2	3	4	5
y		8		3.1	2.4	1.9

(1)

(b) On the grid, draw the graph of $y = \frac{2}{x}\left(5 - \frac{1}{x}\right)$ for $0.5 \leqslant x \leqslant 5$



(2)

(Total for Question 14 is 3 marks)

Answer reference | January 2023 | Question 14

Family: Generate a table and draw a nonlinear graph • Variant: Plot a cubic from a table

14	(a)		12 and 4.5	1	B1 allow $\frac{9}{2}$ oe May be awarded if plotted correctly on the graph
	(b)		Correct graph	2	M1 ft for at least 5 points plotted correctly (± half square)
		Correct answer scores full marks (unless from obvious incorrect working)			A1 for correct curve between $x = 0.5$ and $x = 5$ (clear intention to go through all the points and which must be curved) **Note**: If a fully correct graph is shown, but an incomplete table is shown in (a), then award the marks for (a)
					Total 3 marks

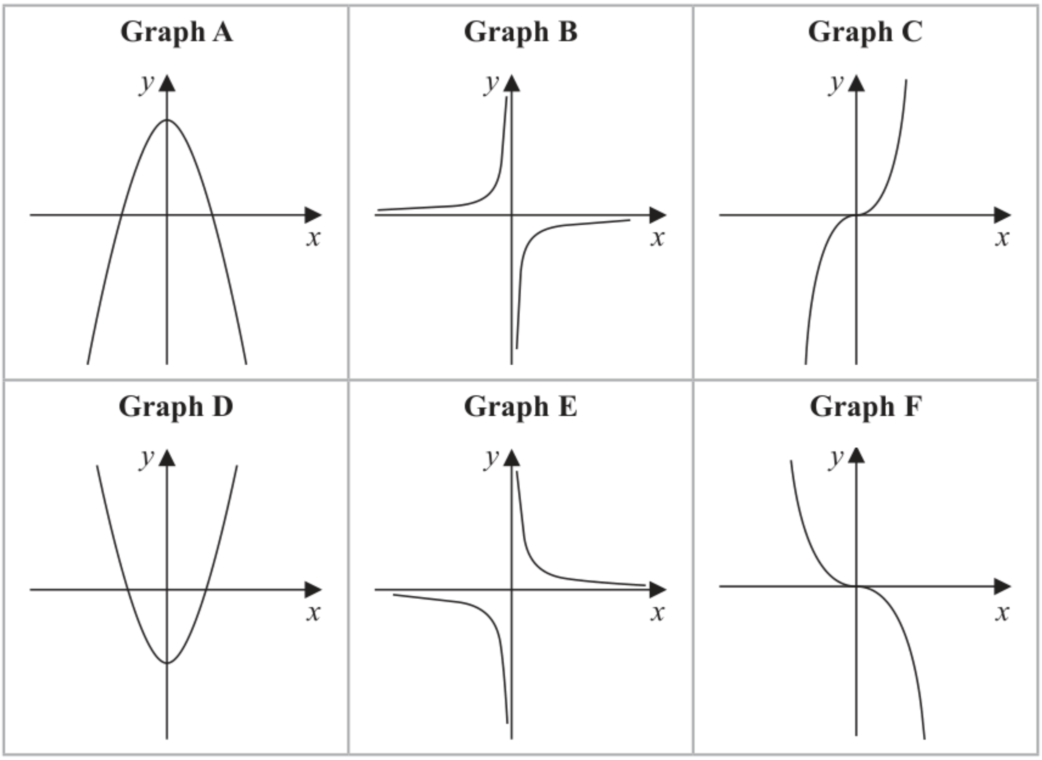
Worked solution

January 2023 | Question 14 | 3 marks

Family: Generate a table and draw a nonlinear graph • Variant: Plot a cubic from a table

Official final mark-scheme pages are embedded immediately after this question.

11 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.
Write your answers on the dotted lines.

Equation	Graph
$y = -\dfrac{2}{x}$	....
$y = 5 - x^2$	....
$y = -2x^3$	....

(Total for Question 11 is 3 marks)

Answer reference | January 2023 | Question 11

Family: Generate a table and draw a nonlinear graph • Variant: Plot a cubic from a table

Question	Working	Answer	Mark	Notes
11		**B**	3	B1
		A		B1
		F		B1
				Total 3 marks

Worked solution

January 2023 | Question 11 | 3 marks

Family: Generate a table and draw a nonlinear graph • Variant: Plot a cubic from a table

Official final mark-scheme pages are embedded immediately after this question.